

☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

20

Q1

ALGEBRA

John paid a total of \$ 165 for a microscope by making a down payment of \$ 37 plus p monthly payments of \$ 16 each. Which of the following equations represents this situation?

- A. $16p - 37 = 165$
- B. $37p - 16 = 165$
- C. $16p + 37 = 165$
- D. $37p + 16 = 165$

Q2

ALGEBRA

$$f(x) = x + b$$

For the linear function f , b is a constant. When $x = 0$, $f(x) = 30$. What is the value of b ?

- A. -30
- B. $-\frac{1}{30}$
- C. $\frac{1}{30}$
- D. 30

Q3

ALGEBRA

Jay walks at a speed of 3 miles per hour and runs at a speed of 5 miles per hour. He walks for w hours and runs for r hours for a combined total of 14 miles. Which equation represents this situation?

A. $3w + 5r = 14$

B. $\frac{1}{3}w + \frac{1}{5}r = 14$

C. $\frac{1}{3}w + \frac{1}{5}r = 112$

D. $3w + 5r = 112$

Q4

ALGEBRA

If $x = 7$, what is the value of $x + 20$?

A. 13

B. 20

C. 27

D. 34

Q5**ALGEBRA**

The function f is defined by $f(x) = \frac{1}{2}x + 6$. What is the value of $f(4)$?

- A. 20
- B. 12
- C. 10
- D. 5

Q6**ALGEBRA**

A line in the xy -plane has a slope of $-\frac{1}{2}$ and passes through the point $(0, 3)$. Which equation represents this line?

- A. $y = -\frac{1}{2}x - 3$
- B. $y = -\frac{1}{2}x + 3$
- C. $y = \frac{1}{2}x - 3$
- D. $y = \frac{1}{2}x + 3$

A gym charges its members a onetime \$ 36 enrollment fee and a membership fee of \$ 19 per month. If there are no charges other than the enrollment fee and the membership fee, after how many months will a member have been charged a total of \$ 188 at the gym?

- A. 4
- B. 5
- C. 8
- D. 10

$$f(x) = 14 + 4x$$

The function f represents the total cost, in dollars, of attending an arcade when x games are played. How many games can be played for a total cost of \$ 58?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

ALGEBRA

A city's total expense budget for one year was x million dollars. The city budgeted y million dollars for departmental expenses and 201 million dollars for all other expenses. Which of the following represents the relationship between x and y in this context?

- A. $x + y = 201$
- B. $x - y = 201$
- C. $2x - y = 201$
- D. $y - x = 201$

Q10

ALGEBRA

3 more than 8 times a number x is equal to 83. Which equation represents this situation?

- A. $(3)(8)x = 83$
- B. $8x = 83 + 3$
- C. $3x + 8 = 83$
- D. $8x + 3 = 83$